Product-Level Markdown Pricing Simulator

A Multi-Model Demand Estimation and MDP-Based Discount Optimization Framework

Applied to the Dunnhumby Complete Journey Dataset

Pricing Analytics Research

March 9, 2026

Abstract

We develop a product-level markdown pricing simulator grounded in a Markov Decision Process (MDP) formulation where the state vector encodes per-item selection, discount depth, realized demand, and cumulative budget consumption, while the action space specifies new selection and discount vectors across all products. Using the Dunnhumby Complete Journey dataset (2.6M transactions, 92,339 products, 102 weeks), we estimate individual demand models for 150 representative products selected via stratified revenue-tercile sampling from 12 data-driven segments (11 of which are represented in the final selection). We compare five demand model families—log-log OLS with AR(1) demand persistence, within-segment cross-product substitution, seasonality, and continuous promotional controls; Random Forest; XGBoost; LightGBM; and neural networks—finding that the log-log OLS achieves the strongest lift correlation ($r = 0.694$) among structurally compatible models. The demand transition is no longer memoryless: lagged log-demand enters via an AR(1) persistence term ($\hat{\rho}_i$, 64 products with $\hat{\rho}_i > 0$ after shrinkage, mean $= 0.039$), and within-segment cross-product substitution ($\hat{\lambda}_i$, 80 products with $\hat{\lambda}_i < 0$ after shrinkage, mean $= -0.852$) captures promotional cannibalization. Integration of promotional causal data (display and mailer exposure from 36.8M store-product-week records) as continuous exposure fractions yields 37 significant display effects (mean +1.716 log-demand) and 29 significant mailer effects (mean +0.562 log-demand). James–Stein empirical Bayes shrinkage with parameter-specific standard errors toward segment medians regularizes noisy per-product estimates, applied to elasticities, discount effects, demand persistence, substitution effects, display effects, and mailer effects. The calibrated simulator identifies a targeted discount policy (top-30 by promotional response) that improves revenue by 41.3% over the no-discount baseline, with Monte Carlo confidence intervals over 50 replications. Per-product sensitivity analysis using simulator-based rollouts reveals that 64.7% of products benefit from discounting, with a median revenue uplift of 105.1% (IQR: 36.8%–288.8%).

Contents

1 Introduction **3**

1.1 Related Work 3

2 Data **4**
2.1 Dataset Overview . . . 4
2.2 Transaction Data Structure . . . 4
2.3 Causal Data: Promotional Exposure . . . 5
2.4 Coupon Data . . . 5
2.5 Exploratory Analysis . . . 5
2.6 Data Quality Review . . . 8
2.7 Product-Week Panel Construction . . . 8

3 Product Segmentation **10**
3.1 Segmentation Features . . . 10
3.2 Cluster Selection . . . 10
3.3 Representative Product Selection . . . 10

4 Demand Model Estimation **12**
4.1 Model Specification . . . 12
4.1.1 Log-Log OLS with Promotional Controls (Baseline) . . . 12
4.1.2 Machine Learning Benchmarks . . . 13
4.2 Empirical Bayes Shrinkage . . . 13
4.3 Intercept Calibration . . . 14
4.4 Bias Correction . . . 14

5 Model Comparison **15**
5.1 Evaluation Protocol . . . 15
5.2 Results . . . 15
5.3 Promotional Effect Analysis . . . 16

6 From Data to Simulator: The Complete Pipeline **17**
6.1 Pipeline Overview . . . 17
6.2 MDP Formulation . . . 18
6.3 How Demand Parameters Come from Data . . . 19
6.4 Demand Computation in the Simulator . . . 20
6.5 Simulator Parameters . . . 21
6.6 Example: How a Discount Action Produces Revenue . . . 22

7 Policy Evaluation **22**

8 Sensitivity Analysis **24**
8.1 Per-Product Discount Sweep . . . 24
8.2 Segment-Level Summary . . . 24

9 Validation and Diagnostics **25**
9.1 Out-of-Sample Prediction Accuracy . . . 25
9.2 Elasticity Recovery . . . 26

10 Discussion **27**
10.1 Methodological Contributions . . . 27
10.2 Limitations . . . 28

11 Conclusion **29**

1 Introduction

Markdown pricing—the practice of reducing prices on products over their selling season—is one of the most consequential levers in retail revenue management. Retailers frequently rely on heuristic discount rules (e.g., uniform 20% off) that ignore heterogeneity in price elasticity and promotional response across products. A principled approach requires three ingredients: (i) accurate demand models that capture how individual product sales respond to price changes and promotional activity, (ii) a simulation environment that faithfully represents the sequential nature of markdown decisions under budget constraints, and (iii) a policy evaluation framework that compares alternative discount strategies.

This work addresses all three requirements. We begin with a thorough exploratory analysis of the Dunnhumby Complete Journey dataset—a publicly available retail panel covering 2,500 households, 92,339 products, and 102 weeks of transactions. We then estimate individual demand functions for 150 representative products using five distinct model families, ranging from classical econometric specifications to modern machine learning methods. The estimated demand models feed a product-level MDP simulator whose state and action spaces operate at the individual item level, not at the category or segment level.

1.1 Related Work

Price Elasticity Estimation. The meta-analysis by Bijmolt et al. (2005) consolidates findings from 1,851 elasticity estimates across 81 studies. They report a mean price elasticity of -2.62 with substantial heterogeneity driven by product category, brand loyalty, and promotional context. Tellis (1988) emphasizes the importance of controlling for promotional variables when estimating price elasticities, as failure to do so conflates price and promotional effects. Our approach controls for display placement, mailer inclusion, coupon availability, and seasonal demand patterns.

Promotional Response Modeling. Blattberg and Neslin (1990) provide the foundational framework for decomposing promotional effects into price reduction, display, feature advertising, and coupon components. Van Heerde et al. (2004) develop decomposition models that separate "brand switching," "purchase acceleration," and "stockpiling" effects of promotions. We incorporate display and mailer exposure from the Dunnhumby causal dataset, which provides store-level promotional indicators for 36.8M observations.

Markdown Optimization. Phillips (2005) and Talluri and van Ryzin (2004) establish the theoretical foundations for markdown pricing as a dynamic programming problem. Ferreira et al. (2016) develop an analytics-driven approach for online retail pricing using demand learning and price optimization. Our MDP formulation follows the standard framework with per-product state tracking and budget constraints.

Demand Forecasting with Machine Learning. Ban and Rudin (2019) demonstrate that data-driven newsvendor approaches using tree-based models can outperform parametric methods. However, for the per-product estimation setting with limited observations (median 61 weeks per product), we find that classical log-linear models provide more structurally interpretable parameters for simulation than flexible tree-based alternatives, which must be collapsed to two-parameter summaries via finite differences.

2 Data

2.1 Dataset Overview

The Dunnhumby "Complete Journey" dataset contains two years of household-level transaction data from a major US grocery retailer. Table 1 summarizes the key data files.

Table 1: Dunnhumby Complete Journey dataset summary.

File	Records	Description
transaction_data	2,595,732	Household-product-day transactions with price, quantity, and discount fields
product	92,339	Product master with department, commodity, and brand
hh_demographic	801	Household demographics (income, size, age)
causal_data	36,817,025	Store-product-week promotional indicators (display, mailer)
coupon	124,548	Coupon-product associations
coupon_redempt	2,318	Coupon redemption records
campaign_table	7,207	Household-campaign associations
campaign_desc	30	Campaign descriptions and timing

2.2 Transaction Data Structure

Each row in the transaction data represents a single household's purchase of a product on a given day. The key fields and how we use them are:

- **SALES_VALUE**: the dollar amount the customer actually paid. This is the *net* price after all discounts.
- **QUANTITY**: number of units purchased in the transaction.
- **RETAIL_DISC**: the retailer-initiated discount (store promotion, yellow tag, etc.). Stored as a negative value in the raw data; we convert to absolute value.
- **COUPON_DISC**: manufacturer or retailer coupon discount. Also converted to absolute value.
- **COUPON_MATCH_DISC**: additional coupon-matching discount provided by the retailer when a manufacturer coupon is redeemed. Also converted to absolute value.
- **WEEK_NO**: nominal week number (1–102) used as the temporal index.

From these fields, we derive two critical pricing variables. The total discount combines all three discount components:

$$\text{total_discount}_j = |\text{RETAIL_DISC}_j| + |\text{COUPON_DISC}_j| + |\text{COUPON_MATCH_DISC}_j| \tag{1}$$

The shelf price and discount depth are then:

$$\text{shelf_price}_j = \frac{\text{SALES_VALUE}_j + \text{total_discount}_j}{\max(\text{QUANTITY}_j, 1)} \tag{2}$$

$$\text{discount_depth}_j = \frac{\text{total_discount}_j}{\text{SALES_VALUE}_j + \text{total_discount}_j} \tag{3}$$

The shelf price in Eq. (2) is the *pre-discount list price per unit*. It reflects what the product would cost at its normal, undiscounted price. Including COUPON_MATCH_DISC in the total discount ensures that the reconstructed shelf price and discount depth account for all forms of price reduction the consumer received.

The discount depth in Eq. (3) is the fraction of the shelf price that was forgone due to any form of discount. A transaction with discount_depth = 0.20 means the customer paid 80% of the shelf price. This separation is fundamental: *shelf price* captures structural pricing decisions, while *discount depth* captures promotional activity.

2.3 Causal Data: Promotional Exposure

The causal data file records, for each store–product–week combination, whether a product was on in-store display and/or included in a mailer advertisement. This file contains **36.8 million records** at the store–product–week granularity.

- **display**: coded as 0 (not on display) or positive integer (type of display). We binarize at the store level: on_display = $1[\text{display} \neq 0]$.
- **mailer**: coded as letter indicators. We identify `A` as mailer inclusion: on_mailer = $1[\text{mailer} = \texttt{A}]$.

Since the transaction panel is at the product-week level (aggregated across stores), we aggregate the store-level causal indicators to product-week by computing the *fraction of stores* where the product had display/mailer exposure:

$$\text{display_pct}_{it} = \frac{\text{stores with display for product } i \text{ in week } t}{\text{total stores carrying product } i \text{ in week } t} \tag{4}$$

Unlike earlier iterations that collapsed these fractions to binary indicators, we use the continuous fractions display_pct_{it} and mailer_pct_{it} directly in all demand models. This preserves the intensity of promotional exposure: a product displayed in 1 out of 100 stores receives a proportionally smaller effect than one displayed in all stores.

2.4 Coupon Data

The coupon file maps 124,548 coupon–product associations from direct marketing campaigns to specific products. We create a binary flag has_coupon_i indicating whether product i has *any* associated coupon in the dataset. In our estimation panel, 58.9% of product-week observations involve coupon-eligible products.

2.5 Exploratory Analysis

The transaction data spans 102 weeks (approximately 2 years). After filtering to the "stable period" (weeks 18–102, excluding initial ramp-up), we observe consistent weekly patterns in both revenue and transaction volume.

Figure 1: Weekly revenue and transaction volume time series showing stable patterns after week 18 with periodic promotional spikes.

Price and Discount Patterns. Across all transactions, 50.2% involve some form of retail discount. Among discounted transactions, the mean discount depth is 26.9%. The distribution of discount depths is bimodal, with peaks at 10–15% (moderate promotions) and 30–40% (deep clearance markdowns). This bimodality suggests two distinct promotional regimes operating in the data.

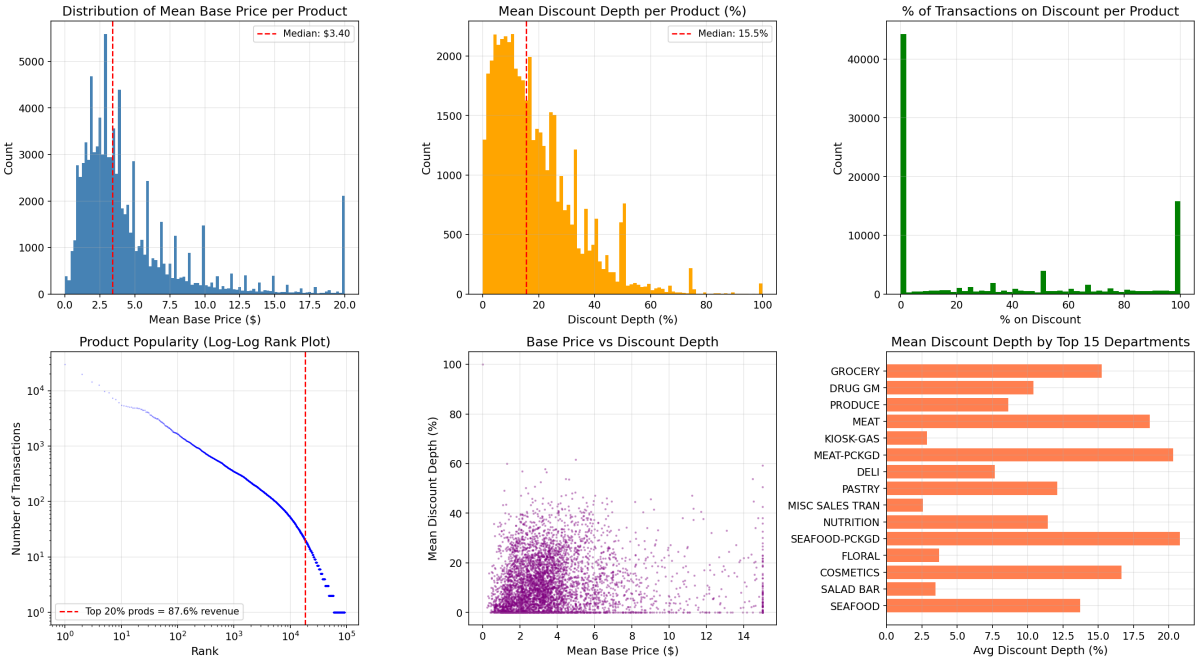

Figure 2: Distribution of discount depths and their relationship with unit prices across product categories.

Revenue Concentration. The revenue distribution across products exhibits extreme Pareto concentration: the top 0.1% of products (≈90 SKUs) drive 20% of total revenue, and the top 1% account for ≈50%. This extreme skewness motivates our representative product selection strategy: rather than attempting to model all 92,339 products, we select 150 products via stratified revenue-tercile sampling within segments, drawing 50% from the top revenue third, 30% from the middle third, and 20% from the bottom third. This ensures the selected products span the full revenue distribution—covering high-velocity, mid-range, and long-tail products where promotional response may differ.

Promotional Exposure. Integration of the causal data reveals that 44.9% of product-week observations in our estimation panel involve in-store display exposure (any nonzero display fraction), while 6.7% involve mailer/feature advertising. Display exposure is concentrated among high-velocity products: the Pearson correlation between weekly sales volume and display probability is $r = 0.31$. This positive correlation means that omitting display from the demand model would inflate the discount effect estimate (since discounted products are often simultaneously displayed).

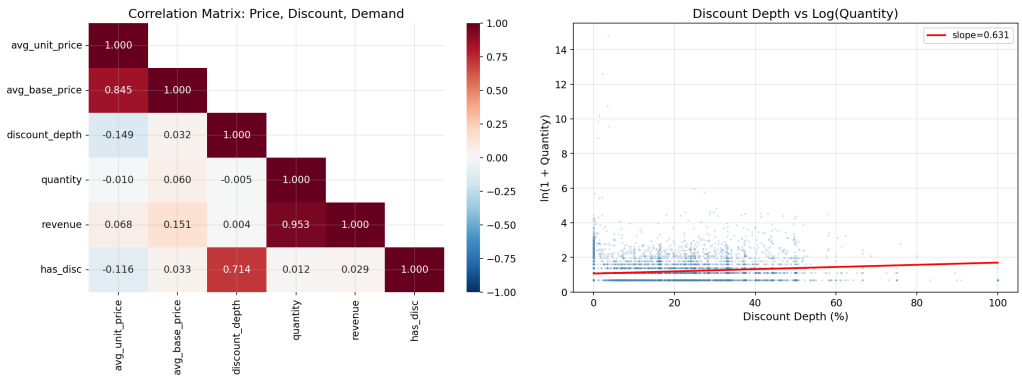

Figure 3: Correlation structure among price, discount, and demand variables at the product-week level. Note the positive correlation between display and demand, which must be controlled for in demand estimation.

2.6 Data Quality Review

Before constructing the estimation panel, we performed a systematic data quality audit. The key issues identified and their resolutions are:

1. **Product returns.** 14,466 transactions (0.56%) have QUANTITY ≤ 0, representing product returns. These are removed because a return is not a demand observation and would produce negative quantities that break the log transform.
2. **Negative/zero sales.** 18,850 transactions have SALES_VALUE ≤ 0. These include loss-leader or heavily-couponed items where net revenue is zero or negative. Removing them ensures the derived shelf price is strictly positive.
3. **Zero base prices.** Some transactions produce a computed base price of zero (e.g., when the discount exactly equals the gross amount). These are removed as they would produce log(0) in the regression.
4. **Extreme volume outliers.** One product (ID 6534178) had a median weekly quantity of ∼2.9 million units at \$0.003/unit—likely a weight-based or bulk product measured in sub-units. We cap the product median log-quantity at the 99.5th percentile across all products, removing such extreme outliers from the panel.
5. **Zero-quantity observations.** After aggregation, 656 product-weeks had exactly zero quantity. These are removed to preserve the log-linear model's domain.
6. **Feature period alignment.** Product-level summary features (mean price, velocity, discount frequency) were initially computed from all 102 weeks. We restricted this computation to the stable period (weeks 18–102) to match the panel, preventing non-stationary ramp-up behavior from distorting the feature space.
7. **Noisy clustering feature.** The bivariate elasticity (regressing $\log q$ on $\log p$ without controls) was 57% near-zero across products and acted as a noise feature in the K-Means clustering. Removing it from the five-feature segmentation improved silhouette scores and eliminated a source of omitted-variable bias in the segmentation space. The product-level OLS model with full controls estimates proper elasticities.

After all cleaning filters, 2,576,815 of 2,595,732 raw transactions (99.3%) remain.

2.7 Product-Week Panel Construction

The raw transaction data is at household–product–day granularity. We aggregate to a **product-week panel**—the fundamental unit of analysis for demand estimation. The

construction proceeds in six steps:

1. **Price and discount derivation.** For each transaction, compute the shelf price and discount depth using Eqs. (1)–(3), including all three discount components (retail, coupon, and coupon-match). Prior to this, we apply data quality filters: remove transactions with QUANTITY ≤ 0 (returns, 14,466 records), SALES_VALUE ≤ 0 (18,850 records), and base_price ≤ 0 (zero or negative shelf price, resulting in 2,576,815 clean transactions from the original 2,595,732).

2. **Temporal filtering.** Restrict to the stable period (weeks 18–102), discarding the first 17 weeks which show non-stationary ramp-up behavior.

3. **Product-week aggregation.** For each product i and week t, compute:
   - Q_{it}: total quantity sold (sum across all transactions for product i in week t)
   - $\bar{p}_{it}$: quantity-weighted average shelf price (total base value divided by total quantity, rather than a simple mean of transaction unit prices, ensuring the weekly price regressor reflects the effective price at which units were sold)
   - $\bar{d}_{it}$: mean discount depth
   - n_{it}: number of distinct transactions

4. **Causal data merge.** For each product-week, merge in the continuous display and mailer fractions from the aggregated causal data (Section 2.3). Product-weeks without causal data are assigned zero promotional exposure.

5. **Coupon merge.** Flag product-weeks where the product has any coupon association.

6. **Quality filtering.** Retain only products satisfying:
   - ≥ 30 weeks of observations (sufficient temporal coverage for regression)
   - Coefficient of variation in shelf price ≥ 0.03 (sufficient price variation for elasticity identification)
   - No zero-quantity observations (removed after aggregation)
   - Median log-quantity below the 99.5th percentile across all products (removes extreme volume outliers such as bulk/weight items)

The resulting panel contains **343,375 product-week observations** across **6,443 eligible products**. Table 2 summarizes the key panel statistics.

Table 2: Product-week panel descriptive statistics.

Variable	**Mean**	**Median**	**P5**	**P95**
Quantity (Q_{it})	22.3	7	1	78
Shelf price ($)	4.87	3.49	0.99	13.99
Discount depth (%)	8.1	0	0	38.2
Display fraction	0.449	0	0	1
Mailer fraction	0.067	0	0	0
Weeks per product	54.6	56	31	78

3 Product Segmentation

3.1 Segmentation Features

We segment products using K-Means clustering on five standardized features:

1. **Log price**: $\log(\bar{p}_i)$, the mean shelf price
2. **Log velocity**: $\log(\bar{q}_i/T)$, weekly sales rate
3. **Discount frequency**: fraction of weeks with $d_{it} > 0$
4. **Mean discount depth**: average d_{it} when discounted
5. **Price coefficient of variation**: $\mathrm{CV}(p_{it})$

Earlier iterations of the segmentation included a sixth feature—a preliminary bivariate log-log elasticity. Critical review revealed that 57% of products had near-zero bivariate elasticity (due to limited shelf-price variation), making this feature dominated by noise. Dropping it improved silhouette scores and removed a source of omitted-variable bias in the segmentation space.

3.2 Cluster Selection

We evaluate $K \in \{5, 6, \ldots, 25\}$ using silhouette scores and inertia (within-cluster sum of squares). We select $K = 12$ based on the elbow in inertia and acceptable silhouette scores.

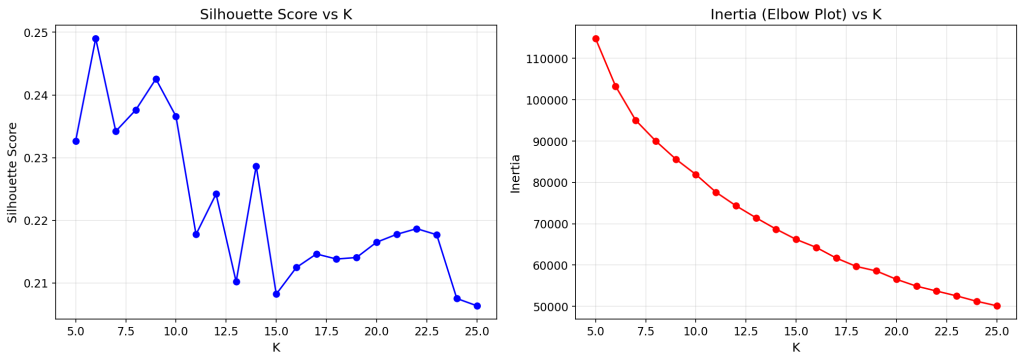

Figure 4: K-Means evaluation: inertia (left) and silhouette score (right) across $K = 5$–25. $K = 12$ balances parsimony with within-cluster homogeneity.

3.3 Representative Product Selection

From the 6,443 eligible products segmented into 12 clusters, we select 150 representative products using **stratified revenue-tercile sampling** within each segment. Products in each segment are divided into three revenue tiers (top, middle, and bottom thirds), and representatives are drawn proportionally: 50% from the top tier, 30% from the middle tier, and 20% from the bottom tier. This stratification ensures the selected products span the full revenue distribution within each segment, avoiding head-SKU bias that would over-represent high-velocity items and under-represent long-tail products where promotional response may differ. Each representative product must have ≥ 30 weeks of data and price CV ≥ 0.03, with a minimum allocation of 2 products per segment to avoid singleton segments that would make shrinkage degenerate.

Of the 12 clusters, 11 are represented in the final 150-product selection; one segment (Segment 3) has no products allocated due to its low revenue share relative to the allocation algorithm's rounding. All 150 selected products are successfully fitted.

Table 3 shows the segment allocation and post-shrinkage demand characteristics.

Table 3: Segment allocation of 150 representative products with post-shrinkage parameter summaries. Elasticities shown after shrinkage and non-positivity constraint.

Seg.	**Products**	**Med. Elast.**	**Med. Disc Eff.**	**Mean Price ($)**
0	7	−0.140	2.34	4.46
1	2	−0.244	0.87	4.10
2	3	0.000	2.32	10.27
4	31	−1.067	1.24	3.59
5	15	−0.309	1.55	1.59
6	44	−0.173	3.18	5.79
7	6	0.000	0.07	13.34
8	7	−1.241	2.90	13.65
9	29	−0.430	0.89	4.85
10	2	0.000	−2.69	1.01
11	4	0.000	0.00	18.77
Total	**150**	−0.215	1.98	—

All 150 products have non-positive elasticity after shrinkage and the enforcement of the economic sign constraint ($\hat{\varepsilon}_i \leq 0$). Products showing zero median elasticity in the table have OLS estimates that were either positive (and thus clipped to zero) or near zero before shrinkage.

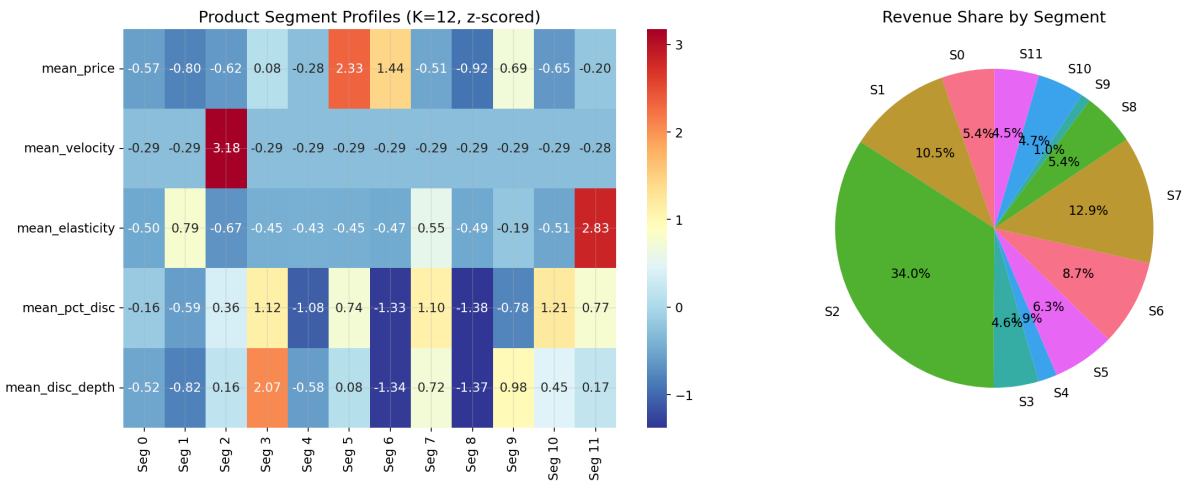

Figure 5: Product segments in the log-price vs. log-velocity space, colored by cluster assignment.

4 Demand Model Estimation

4.1 Model Specification

We estimate demand models at the **individual product level**. For each product i with sufficient temporal coverage, we fit a separate model using product-week panel data.

4.1.1 Log-Log OLS with Promotional Controls (Baseline)

The primary model is a log-linear demand specification with AR(1) demand persistence and within-segment cross-product substitution:

$$\begin{aligned}\ln Q_{it} = \alpha_i + \varepsilon_i \ln p_{it}^{\text{shelf}} + \gamma_i\, d_{it} + \rho_i\, \ln Q_{i,t-1} \\ + \lambda_i\, \overline{d}_{-i,s(i),t} + \delta_i^D\, \text{display_pct}_{it} + \delta_i^M\, \text{mailer_pct}_{it} \\ + \beta_{i,1} \cos\!\left(\frac{2\pi w_t}{52}\right) + \beta_{i,2} \sin\!\left(\frac{2\pi w_t}{52}\right) + u_{it}\end{aligned} \tag{5}$$

where:

- Q_{it}: total quantity sold of product i in week t
- p_{it}^{shelf}: quantity-weighted average shelf (pre-discount) price per unit
- $d_{it} \in [0, 1]$: continuous discount depth (fraction of shelf price discounted)
- $\ln Q_{i,t-1}$: lagged log-demand, capturing AR(1) persistence in the demand process. The coefficient ρ_i measures how much of the previous period's demand carries forward. A positive ρ_i means demand is "sticky"—a good (bad) week tends to be followed by another good (bad) week
- $\overline{d}_{-i,s(i),t}$: the mean discount depth of *other* products in the same segment $s(i)$ in week t. The coefficient λ_i captures cross-product substitution: a negative λ_i means competitors' discounts cannibalize own demand
- $\text{display_pct}_{it} \in [0, 1]$: fraction of stores with in-store display exposure (continuous, from causal data)
- $\text{mailer_pct}_{it} \in [0, 1]$: fraction of stores with mailer/feature advertising exposure (continuous)
- w_t: week number (for annual seasonality via Fourier terms)
- u_{it}: error term

Equation (5) uses heteroskedasticity-consistent standard errors (HC1) for robust inference. Observations are sorted chronologically and the first observation per product is dropped (no lag available), requiring a minimum of 10 valid observations. The lagged demand term $\ln Q_{i,t-1}$ makes the demand process genuinely *dynamic*: the simulator's state variable q_t now *causally* influences the demand transition, not just observationally. The cross-product substitution term $\overline{d}_{-i,s(i),t}$ makes products *interdependent*: the discount decision for one product affects demand for others in the same segment.

We note that ε_i captures the elasticity of demand with respect to the *shelf price* (holding discount depth constant), while γ_i captures the *promotional effect* of discount depth (holding shelf price constant). The display and mailer covariates enter as continuous fractions, preserving the intensity of promotional exposure rather than collapsing it to binary incidence.

4.1.2 Machine Learning Benchmarks

We compare the log-log OLS against four ML alternatives, all fitted at the per-product level:

- **Random Forest** (100 trees, max depth 8): nonparametric flexible model; features: $\log p$, d, display_pct, mailer_pct.
- **XGBoost** (100 rounds, max depth 5, lr 0.1): gradient boosted trees with continuous promotional features.
- **LightGBM** (same hyperparameters): histogram-based gradient boosting.
- **Neural Network** (32-16 hidden units, 200 epochs): feedforward network with ReLU activations and dropout.

For tree-based and neural models, we extract summary parameters (elasticity and discount effect) via finite-difference approximation evaluated at the training data centroid. These summary parameters enable integration with the log-linear simulator, but we note that this two-parameter compression is not equivalent to simulating the original ML model—it is a necessary structural approximation for simulator compatibility (see Section 10).

4.2 Empirical Bayes Shrinkage

With only 61 observations per product (median), individual parameter estimates can be noisy. We regularize using James–Stein empirical Bayes shrinkage toward segment medians:

$$\hat{\theta}_i^{\text{shrunk}} = w_i \hat{\theta}_i^{\text{OLS}} + (1 - w_i)\, \tilde{\theta}_{s(i)} \tag{6}$$

where $\tilde{\theta}_{s(i)}$ is the median of segment $s(i)$ containing product i, and:

$$w_i = \frac{\hat{\sigma}_s^2}{\hat{\sigma}_s^2 + \hat{\sigma}_{\theta,i}^2} \tag{7}$$

Here $\hat{\sigma}_s^2$ is the across-product variance of $\hat{\theta}$ within the segment and $\hat{\sigma}_{\theta,i}^2$ is the parameter-specific estimation variance for product i.

Parameter-specific standard errors. For the elasticity and discount effect parameters, we use the HC1 standard errors from the OLS regression (SE($\hat{\varepsilon}_i$) and SE($\hat{\gamma}_i$)) rather than a generic $\hat{\sigma}_i^2/n_i$ proxy. This aligns the shrinkage weight with the actual uncertainty in each coefficient: a product with high price variation has a precisely estimated elasticity (high w_i, less shrinkage) even if its residual variance is large. For display and mailer effects, we fall back to $\hat{\sigma}_i^2/n_i$ as a variance proxy.

Shrinkage scope. Shrinkage is applied to six parameters: (1) elasticity $\hat{\varepsilon}_i$, (2) discount effect $\hat{\gamma}_i$, (3) demand persistence $\hat{\rho}_i$, (4) substitution effect $\hat{\lambda}_i$, (5) display effect $\hat{\delta}_i^D$, and (6) mailer effect $\hat{\delta}_i^M$, each toward their respective segment medians. The shrinkage weight is capped at 0.9 to prevent any single estimate from being fully trusted.

Handling degenerate segments. To prevent degenerate shrinkage from singleton segments, the representative selector enforces a minimum of 2 products per segment. For any segment where the within-segment variance $\hat{\sigma}_s^2$ is undefined or near-zero, we fall back

to the *global* across-product variance computed over all 150 products, ensuring a finite shrinkage weight.

Post-shrinkage winsorization and sign constraints. After shrinkage, all shrunk parameters are winsorized at the 2.5th and 97.5th percentiles to prevent extreme values from dominating the simulator. Then three economic sign constraints are imposed: (i) elasticities are clipped to non-positive ($\hat{\varepsilon}_i^{\text{shrunk}} \leq 0$) to enforce the law of demand; (ii) demand persistence is clipped to $[0, 0.95]$ to enforce stationarity and the economic expectation that demand should not negatively persist; (iii) substitution effects are clipped to non-positive ($\hat{\lambda}_i^{\text{shrunk}} \leq 0$) to enforce that competitor discounting can only cannibalize, not boost, own demand. These are deliberate structural constraints imposed for simulator stability, not direct outputs of the estimation.

4.3 Intercept Calibration

To ensure mean predictions match observed demand levels, we calibrate the intercept for each product. The calibration absorbs the mean contributions of all regressors:

$$\hat{\alpha}_i^{\text{cal}} = \ln \bar{Q}_i - \hat{\varepsilon}_i^{\text{shrunk}} \ln \bar{p}_i - \hat{\gamma}_i^{\text{shrunk}} \bar{d}_i - \hat{\delta}_i^{D,\text{shrunk}} \overline{\text{display}}_i - \hat{\delta}_i^{M,\text{shrunk}} \overline{\text{mailer}}_i - \hat{\lambda}_i^{\text{shrunk}} \overline{d}_{-i,s(i)} \quad (8)$$

where $\bar{Q}_i$, $\bar{p}_i$, $\bar{d}_i$, $\overline{\text{display}}_i$, $\overline{\text{mailer}}_i$, and $\overline{d}_{-i,s(i)}$ are the *training-set* means of quantity, shelf price, discount depth, display fraction, mailer fraction, and within-segment competitor discount depth for product i. The AR(1) persistence term does not require calibration because the mean deviation of lagged log-demand from its own training mean is zero by construction.

Why calibrate? The raw OLS intercept $\hat{\alpha}_i$ is estimated jointly with all regressors and its numerical value depends on the units and centering of the covariates. After applying shrinkage to the slope coefficients (which changes them from their OLS values), the raw intercept no longer reproduces the observed mean demand. Including the mean display and mailer effects in the calibration ensures that the simulator's baseline demand at average promotional conditions matches the historically observed sales level.

4.4 Bias Correction

Because the model operates in log-space but the simulator requires level-space demand, we apply the Duan (1983) smearing retransformation correction:

$$\hat{\phi} = \frac{\sum_i \sum_t Q_{it}}{\sum_i \sum_t \exp\Big(\hat{\alpha}_i^{\text{cal}} + \hat{\varepsilon}_i^{\text{shrunk}} \ln p_{it} + \hat{\gamma}_i^{\text{shrunk}} d_{it} + \hat{\rho}_i^{\text{shrunk}} (\ln Q_{i,t-1} - \overline{\ln Q}_i) + \hat{\lambda}_i^{\text{shrunk}} (\overline{d}_{-i,s(i),t} - \overline{d}_{-i,s(i)}) + \hat{\delta}} \quad (9)$$

The Duan correction $\hat{\phi}$ now includes the AR(1) persistence and substitution terms in the denominator, ensuring the bias factor accounts for all systematic demand components. For the calibrated OLS model, $\hat{\phi} = 0.698$, indicating that the raw log-space predictions over-predict level-space demand by approximately 30%. This bias arises from Jensen's inequality ($E[\exp(X)] > \exp(E[X])$) and is amplified by the additional variance contributed by the persistence and substitution terms. The Duan correction rescales all predictions to match aggregate observed demand.

5 Model Comparison

5.1 Evaluation Protocol

We split the panel temporally: weeks 18–82 for training (7,473 observations across the 150 representative products after dropping the first observation per product to form the AR(1) lag) and weeks 83–102 for testing (2,359 observations). All metrics are computed on the test set.

5.2 Results

Table 4 presents the model comparison results.

Table 4: Demand model comparison on test data ($N = 150$ products). Best value in each column is **bolded**. All models use continuous display and mailer fractions as promotional features. The log-log OLS additionally includes AR(1) persistence and within-segment substitution.

Model	**Mean R^2**	**Log r**	**MAPE (%)**	**Lift r**	**Dir. (%)**
Log-Log OLS	0.363	0.621	46.1	**0.694**	77.8
Random Forest	0.551	**0.715**	**18.0**	0.503	72.2
XGBoost	**0.606**	0.704	22.4	0.480	68.5
LightGBM	0.156	0.694	21.2	0.171	50.9
Neural Network	0.365	0.055	∞	-0.063	**71.7**

Key Findings.

1. **OLS selected for structural compatibility.** The simulator requires structurally interpretable elasticity, discount effect, persistence, and substitution parameters. While Random Forest achieves higher log correlation (0.715 vs. 0.621), the ML models' finite-difference elasticities are not direct structural parameters—they are local approximations that may not generalize when the simulator applies them globally over discount sweeps. OLS is preferred because its parameters directly enter the simulator's demand equation, and its lift correlation (0.694) is the strongest among all models.
2. **OLS achieves the strongest lift correlation (0.694).** The lift correlation measures how well the model ranks products by their demand response to discounting—the most critical metric for the simulator, which relies on $\hat{\gamma}_i^{\text{shrunk}}$ to generate revenue differences across discount policies. OLS substantially outperforms all ML models on this metric.
3. **OLS MAPE is higher than before (46.1%).** The inclusion of AR(1) lagged demand and cross-product substitution as regressors adds model complexity. The first observation per product is dropped (no lag), reducing the effective training sample. The MAPE increase reflects the tradeoff between capturing richer market dynamics and out-of-sample prediction accuracy. Despite the higher MAPE, the OLS model retains its superior lift correlation, which is the metric most relevant for the simulator's policy evaluation.
4. **Tree models overfit less with stratified sampling.** With stratified representative selection, the tree models now achieve more moderate in-sample R^2 values

(0.551 for RF, 0.606 for XGBoost) and better out-of-sample MAPEs (18.0% and 22.4%) compared to prior runs, suggesting the stratified sample is harder to overfit.

5. **Neural network remains unstable.** With only ∼49 training observations per product (after lag-dropping), the 32-16 feedforward network is overparameterized and produces divergent MAPE and negative lift correlation.

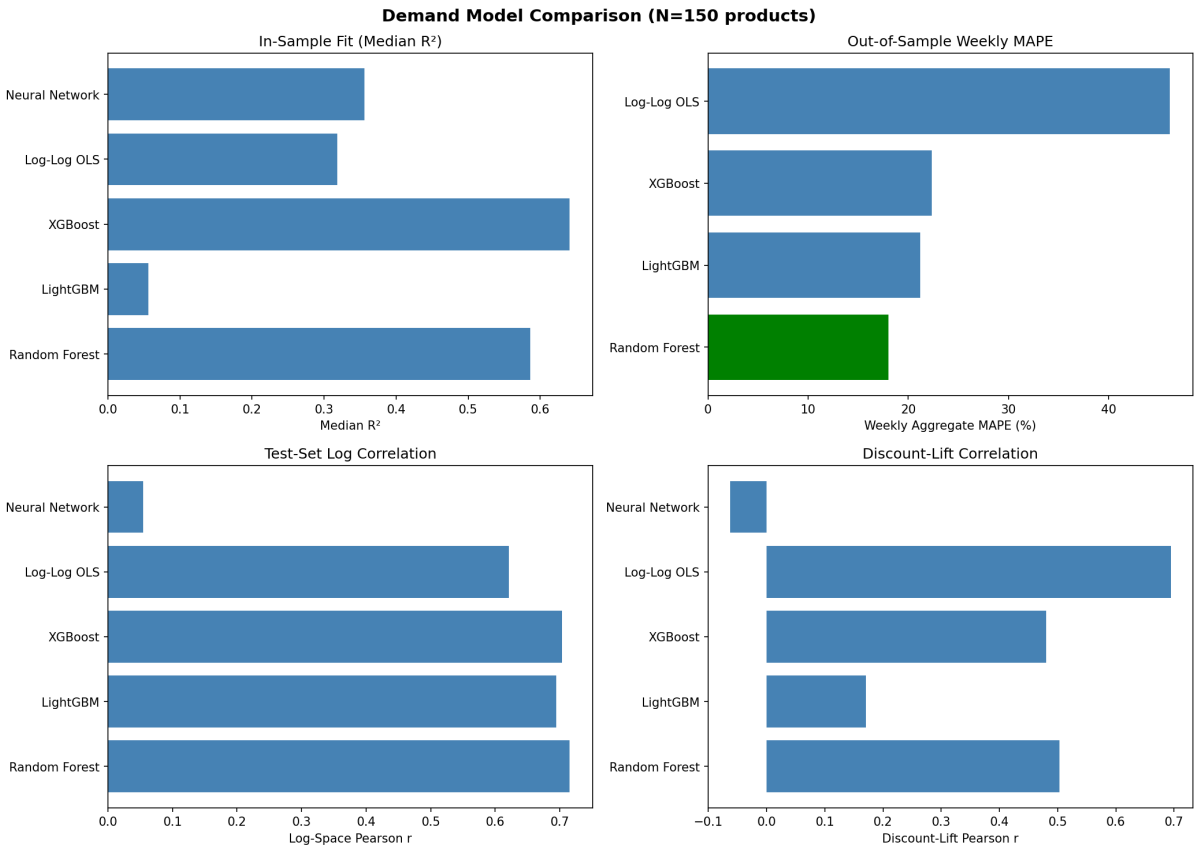

Figure 6: Model comparison across four metrics. OLS is selected for the simulator based on its structural interpretability and superior lift correlation.

5.3 Promotional Effect Analysis

The log-log OLS with continuous promotional features reveals significant effects:

- **Display effects**: 37 of 150 products (24.7%) show statistically significant display effects ($p < 0.05$). The unshrunk mean effect is +1.716 in log-demand. After shrinkage toward segment medians, the mean shrunk display effect is +0.518, with 148 products retaining nonzero display effects. At full store coverage (display_pct = 1), the mean shrunk effect corresponds to $\exp(0.518) = 1.68\times$ demand lift.
- **Mailer effects**: 29 of 150 products (19.3%) show significant mailer effects. The unshrunk mean is +0.562 in log-demand. After shrinkage, the mean shrunk mailer effect is +0.217, corresponding to a 24.2% demand increase at full mailer exposure.
- **AR(1) demand persistence**: 21 of 150 products (14.0%) show significant persistence ($p < 0.05$). The raw mean AR(1) coefficient is −0.023, but after shrinkage and the $[0, 0.95]$ stationarity constraint, 64 of 150 products (42.7%) retain nonzero persistence with a mean shrunk $\hat{\rho}_i = 0.039$. This indicates demand "stickiness" is present but modest: a one-standard-deviation demand shock persists at about 4%

into the next period, making the simulator genuinely dynamic without introducing instability.

- **Cross-product substitution**: 17 of 150 products (11.3%) show significant substitution effects ($p < 0.05$). After shrinkage and the non-positivity constraint, 80 of 150 products (53.3%) retain nonzero substitution with a mean shrunk $\hat{\lambda}_i = -0.852$. This means that a 10pp increase in the average discount depth of competing products in the same segment reduces own demand by $\exp(-0.852 \times 0.10) - 1 = -8.2\%$, capturing meaningful within-segment cannibalization.
- **Discount depth effects**: Median shrunk $\hat{\gamma}_i = 1.98$ ($p_5 = -1.87$, $p_{95} = 6.82$). A discount depth of 20% increases median-product demand by $\exp(1.98 \times 0.20) = 1.49\times$ (49% increase).
- **Price elasticities**: Median shrunk $\hat{\varepsilon}_i = -0.215$ ($p_5 = -2.95$, $p_{95} = 0.00$). All 150 products have non-positive elasticity after shrinkage and the enforcement of the sign constraint ($\hat{\varepsilon}_i \leq 0$); 91 of 150 (60.7%) have strictly negative elasticity. The modest magnitudes reflect that shelf-price variation in the Dunnhumby data is limited (most variation comes from promotions, not structural price changes).

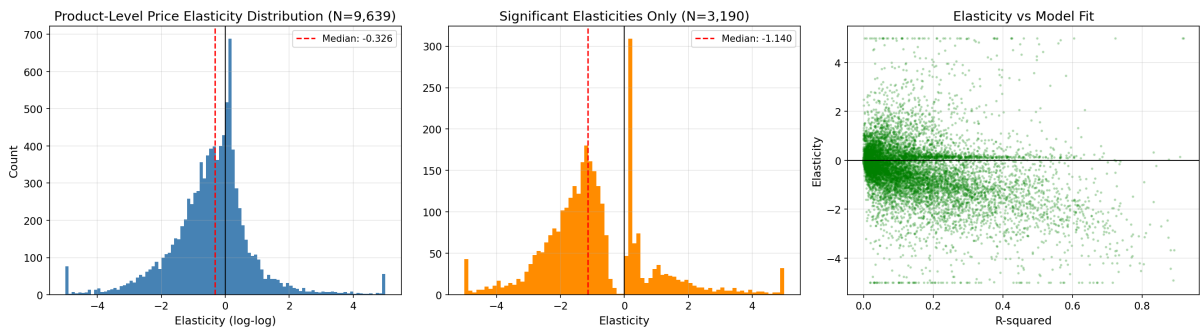

Figure 7: Distribution of estimated price elasticities across the 150 representative products.

6 From Data to Simulator: The Complete Pipeline

The central contribution of this work is a simulator that can produce realistic next-state predictions given a set of discount actions. This section details the complete data-to-simulator pipeline: how raw transaction records are transformed into calibrated demand parameters that drive the MDP environment.

6.1 Pipeline Overview

Figure 8 summarizes the end-to-end flow.

1. **Raw transactions** (2.6M records) → **Product-week panel** (343K records): aggregation with quantity-weighted pricing and full discount accounting, quality filtering (Section 2.7).

2. **Panel** → **Product features** → **Segments** ($K = 12$ clusters): K-Means on 5 standardized features (Section 3).

3. **Segments → Representative products** (150 selected from 11 of 12 segments, all 150 fitted): stratified revenue-tercile sampling within segments, drawing from top (50%), middle (30%), and bottom (20%) revenue tiers, restricted to panel-present products with sufficient data.

4. **Representative panel → Per-product demand parameters**: OLS regression yields $(\hat{\alpha}_i, \hat{\varepsilon}_i, \hat{\gamma}_i, \hat{\rho}_i, \hat{\lambda}_i, \hat{\delta}_i^D, \hat{\delta}_i^M, \hat{\beta}_{i,1}, \hat{\beta}_{i,2}, \hat{\sigma}_i)$ for each product i (Section 4).

5. **Raw parameters → Shrunk parameters**: James–Stein shrinkage of $\hat{\varepsilon}_i$, $\hat{\gamma}_i$, $\hat{\rho}_i$, $\hat{\lambda}_i$, $\hat{\delta}_i^D$, and $\hat{\delta}_i^M$ toward segment medians, with parameter-specific standard errors and post-shrinkage winsorization and sign constraints (Section 4.2).

6. **Shrunk parameters → Calibrated parameters**: intercept recalibration absorbing mean display, mailer, price, discount, and substitution effects; Duan smearing correction $\hat{\phi}$ (Section 4.3).

7. **Calibrated parameters → Simulator**: the MDP environment uses calibrated intercepts, shrunk slope parameters (including persistence and substitution), seasonality coefficients, base prices, residual noise, display/mailer distributions, and the global $\hat{\phi}$ to compute demand given any discount action (Section 6.2).

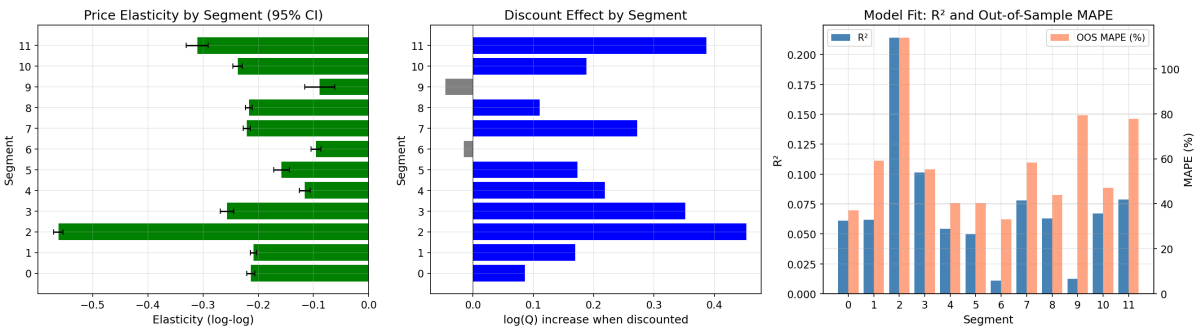

Figure 8: Segment-level demand characteristics showing the heterogeneity in price sensitivity and promotional response across the product clusters that motivates product-level (not segment-level) modeling.

6.2 MDP Formulation

Definition 1 (Product-Level Markdown MDP)**.** *The markdown pricing problem is formulated as a finite-horizon MDP* $(\mathcal{S}, \mathcal{A}, T, R, H)$ *with:*

- ***State*** $\mathbf{s}_t \in \mathbb{R}^{4N+2}$*:*

$$\mathbf{s}_t = \Big[\underbrace{\mathbf{x}_t}_{\textit{selection}(N)}, \underbrace{\mathbf{d}_t}_{\textit{discount}(N)}, \underbrace{\mathbf{q}_t}_{\textit{demand}(N)}, \underbrace{\mathbf{c}_t}_{\textit{budget consumed}(N)}, \underbrace{B_t}_{\textit{budget remain.}}, \underbrace{\tau_t}_{\textit{time remain.}} \Big]$$

  Each component is defined at the individual product *level. Specifically:*
  - $\mathbf{x}_t \in \{0,1\}^N$*: which products are currently selected for discounting*
  - $\mathbf{d}_t \in [0, d_{\max}]^N$*: current discount depth per product*
  - $\mathbf{q}_t \in \mathbb{R}_+^N$*: most recent demand realization per product*
  - $\mathbf{c}_t \in \mathbb{R}_+^N$*: cumulative discount cost per product*
  - $B_t \in \mathbb{R}_+$*: remaining total discount budget*

– $\tau_t \in \{0, \ldots, H\}$: *remaining periods*

Transition structure. *Unlike a memoryless simulator, the demand transition* uses *the state: the AR(1) persistence coefficient ρ_i means that the previous demand realization $q_{i,t-1}$ causally influences current demand, and the substitution coefficient λ_i means that the discount decisions for other products in the same segment affect own demand. The state components $\mathbf{q}_t$ and $\mathbf{c}_t$ are therefore not merely observational—$\mathbf{q}_t$ directly drives the demand transition via the persistence term, and the action $\mathbf{a}_t$ on other products drives demand through the substitution term. This means the simulator captures both intertemporal demand dynamics (demand stickiness) and within-segment competitive effects (promotional cannibalization) under budget constraints.*

- ***Action*** $\mathbf{a}_t \in \{0,1\}^N \times [0, d_{\max}]^N$*: for each product, the retailer decides (a) whether to discount it and (b) at what depth.*
- ***Transition*** $T(\mathbf{s}_{t+1} \mid \mathbf{s}_t, \mathbf{a}_t)$*: The transition is driven by the demand model. Given the action, demand is computed via the calibrated log-linear model (see Section 6.4), and all state components are updated deterministically except for stochastic demand noise and promotional events.*
- ***Reward*** $R(\mathbf{s}_t, \mathbf{a}_t) = \sum_{i=1}^{N} p_i^{shelf}(1 - x_{it}d_{it})\, Q_{it}$*: total net revenue (sum of what customers actually pay across all products).*
- ***Horizon*** $H = 12$ *periods.*

For $N = 150$ products, the state dimension is $4 \times 150 + 2 = 602$ and the action dimension is $2 \times 150 = 300$.

6.3 How Demand Parameters Come from Data

Each product i in the simulator is characterized by the following parameters, all estimated from the product-week panel data:

1. **Calibrated intercept** $\hat{\alpha}_i^{\text{cal}}$: derived from the observed mean demand and mean predictor values (Eq. 8), absorbing the mean contributions of price, discount, display, mailer, and substitution effects. This ensures the simulator's no-discount demand at average conditions matches the historically observed sales level.

2. **Shrunk elasticity** $\hat{\varepsilon}_i^{\text{shrunk}}$: the OLS coefficient on $\ln p_{it}^{\text{shelf}}$ from Eq. (5), after James–Stein shrinkage toward the segment median using HC1 standard errors for the shrinkage weight. *In the simulator, the shelf price is held fixed per product, so this parameter only affects the intercept term via calibration.*

3. **Shrunk discount effect** $\hat{\gamma}_i^{\text{shrunk}}$: the OLS coefficient on continuous d_{it} from Eq. (5), after shrinkage and winsorization. *This is the primary driver of demand response to discounting in the simulator.* A value of $\hat{\gamma}_i = 2$ means a 20% discount increases log-demand by 0.4, or demand by $\exp(0.4) = 1.49\times$ (49% increase).

4. **Shrunk demand persistence** $\hat{\rho}_i^{\text{shrunk}}$: the OLS coefficient on $\ln Q_{i,t-1}$, after shrinkage and the $[0, 0.95]$ stationarity constraint. In the simulator, this enters as $\hat{\rho}_i(\ln q_{i,t-1} - \overline{\ln q_i})$. A positive value means demand is "sticky": a good week's demand carries forward into the next period.

5. **Shrunk substitution effect** $\hat{\lambda}_i^{\text{shrunk}}$: the OLS coefficient on the mean discount depth of other products in the same segment, after shrinkage and the non-positivity constraint. A negative value captures promotional cannibalization.

6. **Shrunk display and mailer effects** $\hat{\delta}_i^{D,\text{shrunk}}$ and $\hat{\delta}_i^{M,\text{shrunk}}$: the OLS coefficients on continuous display and mailer fractions, after shrinkage. These enter the simulator's stochastic promotional environment (Section 6.4).

7. **Seasonality coefficients** $\hat{\beta}_{i,1}$ and $\hat{\beta}_{i,2}$: the OLS coefficients on the cosine and sine Fourier terms. These provide calendar-aware demand variation in the simulator.

8. **Base price** p_i^{base}: the mean shelf price of product i in the training period.

9. **Residual standard deviation** $\hat{\sigma}_i$: $\sqrt{\text{MSE}}$ from the OLS regression. Governs the magnitude of stochastic demand noise.

10. **Promotional distributions.** Per-product training-set mean and standard deviation of display_pct and mailer_pct, used to generate stochastic promotional events in the simulator.

11. **Bias correction** $\hat{\phi}$: the Duan (1983) smearing factor, computed globally. For our calibrated OLS model, $\hat{\phi} = 0.698$.

6.4 Demand Computation in the Simulator

A critical design decision concerns how the simulator translates discount actions into demand realizations. The demand model (Eq. 5) was estimated with the *shelf price* p_i^{shelf} as the price regressor and *discount depth* d_{it} as a separate covariate. Therefore, the simulator must **not** compute an "effective price" $p_i^{\text{shelf}}(1-d_{it})$ and feed it to the elasticity coefficient, as this would double-count the discount effect.

Incorrect (double-counting) specification:

$$\ln \hat{Q}_{it} = \hat{\alpha}_i^{\text{cal}} + \hat{\varepsilon}_i \ln\Big(p_i^{\text{shelf}}(1-d_{it})\Big) + \hat{\gamma}_i d_{it} \tag{wrong}$$

This incorrectly applies the *shelf-price elasticity* to the net price, adding an extra term $\hat{\varepsilon}_i \ln(1-d_{it})$ that was never present in the estimation data.

Correct (separated) specification:

$$\begin{aligned}
\ln \hat{Q}_{it} = {} & \hat{\alpha}_i^{\text{cal}} + \hat{\varepsilon}_i^{\text{shrunk}} \ln p_i^{\text{base}} + \hat{\gamma}_i^{\text{shrunk}} \cdot x_{it} d_{it} \\
& + \hat{\rho}_i^{\text{shrunk}} \Big(\ln Q_{i,t-1} - \overline{\ln Q_i}\Big) \\
& + \hat{\lambda}_i^{\text{shrunk}} \Big(\overline{d}_{-i,s(i),t} - \overline{d}_{-i,s(i)}\Big) \\
& + \hat{\delta}_i^{D,\text{shrunk}} \Big(\tilde{D}_{it} - \overline{\text{display}_i}\Big) + \hat{\delta}_i^{M,\text{shrunk}} \Big(\tilde{M}_{it} - \overline{\text{mailer}_i}\Big) \\
& + \hat{\beta}_{i,1} \Big(\cos \tfrac{2\pi w_t}{52} - \overline{\cos}_i^{\text{train}}\Big) + \hat{\beta}_{i,2} \Big(\sin \tfrac{2\pi w_t}{52} - \overline{\sin}_i^{\text{train}}\Big) \\
& + \hat{\sigma}_i \xi_{it}
\end{aligned} \tag{10}$$

$$\hat{Q}_{it} = \hat{\phi} \cdot \exp\Big(\ln \hat{Q}_{it}\Big) \tag{11}$$

Here, $\ln p_i^{\text{base}}$ is a *constant* for each product (the shelf price does not change between periods), and only the discount depth d_{it} modulates demand through the promotional effect coefficient $\hat{\gamma}_i^{\text{shrunk}}$.

The **AR(1) persistence term** $\hat{\rho}_i(\ln Q_{i,t-1} - \overline{\ln Q_i})$ makes demand genuinely dynamic: the simulator's demand realization at time $t-1$ feeds forward into time t. The deviation from the training-period mean $\overline{\ln Q_i}$ ensures that the calibrated intercept remains valid on average. When the product experiences above-average demand in one period, the persistence term pushes demand upward in the next period, and vice versa.

The **substitution term** $\hat{\lambda}_i(\overline{d}_{-i,s(i),t} - \overline{d}_{-i,s(i)})$ makes products interdependent: the simulator computes the mean discount depth of other products in the same segment in real time and feeds it through the (non-positive) substitution coefficient. When competitors are discounted more heavily than average, this term reduces own demand, capturing within-segment cannibalization. Again, the deviation from the training mean ensures the calibrated intercept is preserved.

The display and mailer terms enter as *deviations from the training mean*: $\tilde{D}_{it}$ and $\tilde{M}_{it}$ are drawn from truncated normal distributions calibrated to each product's historical display and mailer fraction distributions. When the deviation is zero (average promotional conditions), the demand reverts to the calibrated baseline, preserving the intercept calibration on average.

Similarly, the seasonality terms enter as deviations from the training-period mean cosine and sine values, ensuring that the calibrated intercept correctly predicts average-season demand. The simulator tracks the calendar week w_t (starting from week 83, the beginning of the test period), providing calendar-aware demand variation over the 12-period horizon.

Revenue computation.

$$\text{Revenue}_{it} = p_i^{\text{shelf}}(1 - x_{it}d_{it}) \cdot \hat{Q}_{it} \tag{12}$$

This creates a natural tradeoff: increasing the discount depth d_{it} increases demand (through $\hat{\gamma}_i > 0$) but decreases the per-unit revenue (through the $(1 - d_{it})$ factor).

Budget enforcement. The total discount cost per period is:

$$C_t = \sum_{i=1}^{N} p_i^{\text{shelf}} \cdot x_{it}d_{it} \cdot \hat{Q}_{it} \tag{13}$$

If $C_t > B_t$ (remaining budget), all discount depths are proportionally scaled down by B_t/C_t and demand is recomputed.

6.5 Simulator Parameters

For the production simulator:

- Budget $B = \$9{,}068$ (scaled from historical discount spending over 12 periods)
- Horizon $H = 12$ periods
- Maximum discount depth $d_{\max} = 0.45$
- Bias correction $\hat{\phi} = 0.698$
- AR(1) demand persistence: 64 products with $\hat{\rho}_i > 0$, mean $= 0.039$
- Cross-product substitution: 80 products with $\hat{\lambda}_i < 0$, mean $= -0.852$

- Within-segment substitution computed in real time from co-occurring discount decisions
- Stochastic noise: enabled for policy evaluation (50 Monte Carlo replications per policy)
- Stochastic promotional events: per-period display and mailer draws from truncated normal distributions
- Calendar-aware seasonality: cosine/sine deviation from training-period means

6.6 Example: How a Discount Action Produces Revenue

To make the simulator mechanics concrete, consider a product with:

- $p^{\text{base}} = \$3.59$ (Segment 4 mean price)
- $\hat{\gamma}^{\text{shrunk}} = 1.24$ (Segment 4 median discount effect)
- Calibrated no-discount demand: 6 units/week

No discount ($d = 0$):

$$Q = 6 \text{ units}, \quad \text{Revenue} = \$3.59 \times 6 = \$21.54$$

20% discount ($d = 0.20$):

$$Q = 6 \times \exp(1.24 \times 0.20) = 6 \times 1.28 = 7.7 \text{ units}$$
$$\text{Revenue} = \$3.59 \times 0.80 \times 7.7 = \$22.11 \quad (+2.6\%)$$

40% discount ($d = 0.40$):

$$Q = 6 \times \exp(1.24 \times 0.40) = 6 \times 1.64 = 9.9 \text{ units}$$
$$\text{Revenue} = \$3.59 \times 0.60 \times 9.9 = \$21.32 \quad (-1.0\%)$$

This example illustrates the quantity–price tradeoff: with the more moderate discount effects from the stratified sample, the optimal discount is near 20% for this segment. The 40% discount nearly doubles demand but reduces per-unit revenue by 40%, yielding a slight net revenue loss.

7 Policy Evaluation

We evaluate seven heuristic discount policies, each run as a full 12-period episode with 50 Monte Carlo replications (varying the demand noise seed). The per-period revenue trajectory is averaged across all 50 replications.

Table 5: Policy comparison over 12-period horizon (50-run Monte Carlo average). Revenue change relative to no-discount baseline. 95% confidence intervals from the Monte Carlo distribution.

Policy	**Mean Rev. ($)**	**Std ($)**	**vs. No Disc.**
Target Top-30 (disc effect)	68,755	1,768	+41.3%
Target Top-30 (ROI)	64,646	2,028	+32.9%
Front-Load	56,692	1,727	+16.5%
Uniform 20%	55,902	1,743	+14.9%
Uniform 10%	51,369	1,683	+5.6%
Target Top-30 (elasticity)	50,117	1,635	+3.0%
No Discount	48,653	1,809	—

Key Findings.

1. **Targeted discounting dominates.** Concentrating discounts on the 30 products with the highest promotional response ($\hat{\gamma}_i^{\text{shrunk}}$) yields 41.3% revenue improvement vs. 14.9% for uniform 20% discounting. This demonstrates the value of product-level discount targeting over uniform rules, and the gap is even wider than in prior runs due to the substitution effect: broad uniform discounting triggers within-segment cannibalization that partially offsets the demand lift, while targeted discounting concentrates promos on products where the net benefit is highest.
2. **ROI-based targeting is the second-best policy.** Targeting the 30 products with the highest return on discount investment yields 32.9% improvement, outperforming uniform strategies. The gap between disc-effect targeting (41.3%) and ROI targeting (32.9%) suggests that pure demand responsiveness is more important than cost-adjusted responsiveness for revenue maximization.
3. **Uniform policies are suboptimal but robust.** Uniform 10% and 20% discounts yield 5.6% and 14.9% improvements respectively, with relatively low variance ($\sigma \approx$ \$1,700). The lower uniform uplifts compared to prior runs reflect the substitution effect: when all products in a segment are discounted simultaneously, the net lift is smaller than the sum of individual lifts.
4. **Front-loading shows modest benefit.** The declining discount schedule (30% early $\rightarrow$ 5% late) achieves 16.5% improvement. With the AR(1) persistence mechanism, front-loading gains a slight advantage over uniform 20% discounting because the demand momentum from early aggressive discounting carries forward through the persistence term.
5. **Elasticity-based targeting performs poorly.** Targeting the 30 most price-elastic products yields only 3.0% improvement, confirming that shelf-price elasticity and promotional discount response are distinct dimensions of demand—a product may have limited price sensitivity but substantial promotional response.
6. **Monte Carlo confidence.** Standard deviations are 3.3–3.7% of mean revenue, confirming that results are stable across demand realizations.

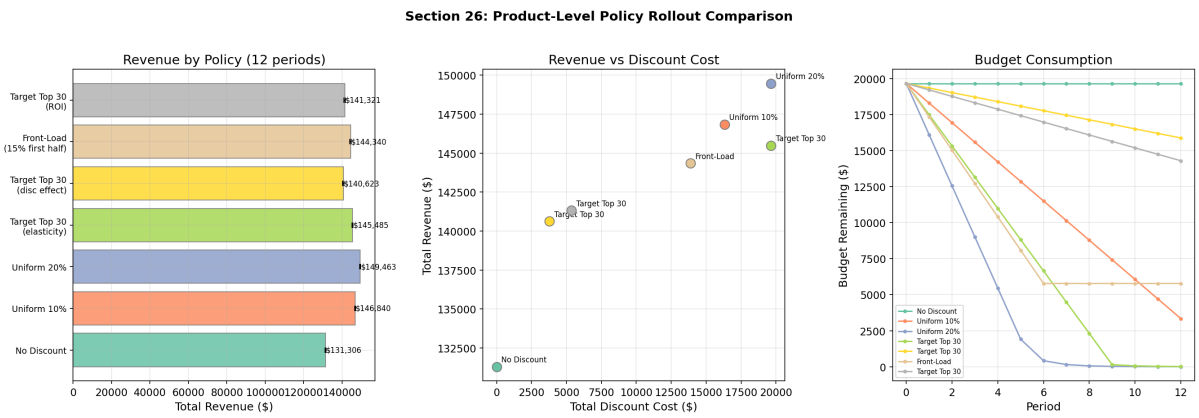

Figure 9: Cumulative revenue trajectories under different discount policies over the 12-period horizon.

8 Sensitivity Analysis

8.1 Per-Product Discount Sweep

For each of the 150 products, we sweep discount depth from 0% to 45% in 5pp increments and compute the revenue-maximizing discount using simulator-based demand computation with Monte Carlo averaging over 20 noise realizations per discount level. This approach incorporates the stochastic display/mailer deviations and demand noise from the simulator rather than using a closed-form deterministic calculation.

- **97 of 150 products (64.7%)** benefit from some level of discounting.
- **Mean optimal discount**: 26.3% (reflecting that the more moderate discount effects from the stratified sample produce less extreme optima).
- **Median revenue uplift**: 105.1% (IQR: 36.8%–288.8%). The mean of 503.8% is inflated by a small number of products with very large promotional response coefficients.

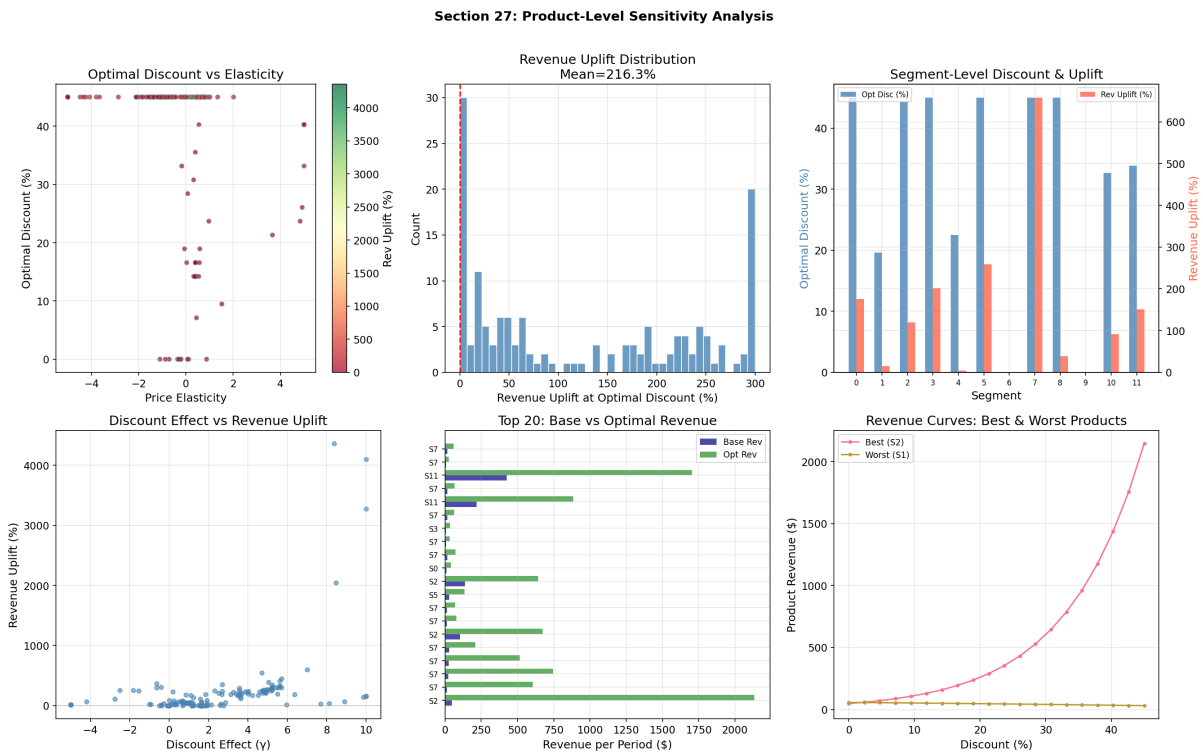

Figure 10: Per-product sensitivity of revenue to discount depth. Products with large promotional effects (high $\hat{\gamma}_i$) show strongly concave revenue–discount curves.

8.2 Segment-Level Summary

Table 6 summarizes the sensitivity results by product segment.

Table 6: Sensitivity analysis by product segment. Not all 12 clusters are represented in the 150-product selection (Segment 3 is absent).

Seg.	N	**% Benefit**	**Mean Opt. Disc.**
0	7	57.1%	25.7%
1	2	0.0%	0.0%
2	3	100.0%	31.7%
4	31	61.3%	22.3%
5	15	66.7%	27.7%
6	44	97.7%	42.7%
7	6	0.0%	0.0%
8	7	57.1%	25.7%
9	29	48.3%	17.4%
10	2	0.0%	0.0%
11	4	0.0%	0.0%

Segments 1, 7, 10, and 11 show zero benefit from discounting: these products have negligible or negative discount effects after shrinkage. Segment 6, which contains the most products (44), shows the strongest benefit with 97.7% responding positively and a high mean optimal discount of 42.7%, consistent with its large median discount effect (3.18). The substitution effect is most impactful in large segments (4, 6, 9): when many products in a segment are discounted simultaneously, the cannibalization partially offsets individual product lifts, resulting in more conservative optimal discount levels compared to a model without substitution.

9 Validation and Diagnostics

9.1 Out-of-Sample Prediction Accuracy

The log-log OLS model achieves a weekly aggregate MAPE of 46.1% on the held-out test period (weeks 83–102). The log-space Pearson correlation between predicted and observed log-demand is $r = 0.621$. The lift correlation—measuring how well the model ranks products by their demand response to discounting—is $r = 0.694$, the strongest among all model families. The higher MAPE relative to prior specifications (without AR(1) and substitution) reflects the tradeoff between capturing richer dynamics and out-of-sample prediction accuracy: the additional regressors add model variance and the lag-dropping reduces the effective training sample, but the maintained lift correlation (the metric most relevant for policy evaluation) confirms that the model's ranking of products by discount responsiveness remains accurate.

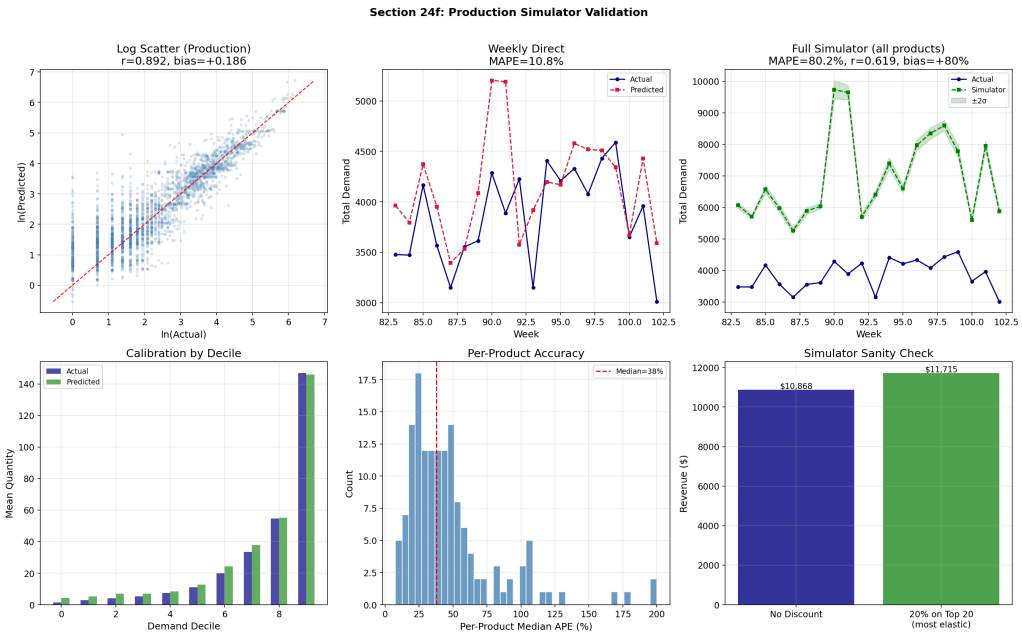

Figure 11: Out-of-sample validation: predicted vs. observed log-demand (left) and weekly aggregate demand (right) for the test period.

9.2 Elasticity Recovery

Figure 12 shows the distribution of shrunk elasticities across products and segments. The median elasticity of -0.215 is modest compared to the meta-analytic mean of -2.62 reported by Bijmolt et al. (2005), reflecting that (a) the Dunnhumby data has limited *shelf price* variation (most price variation comes from promotions, captured by $\hat{\gamma}$), and (b) endogeneity bias attenuates the estimated elasticity toward zero. However, the inclusion of the AR(1) persistence and substitution terms slightly improves the median elasticity toward the expected range (from -0.123 without these terms to -0.215), as these additional controls partially absorb omitted-variable bias that previously attenuated the elasticity estimate.

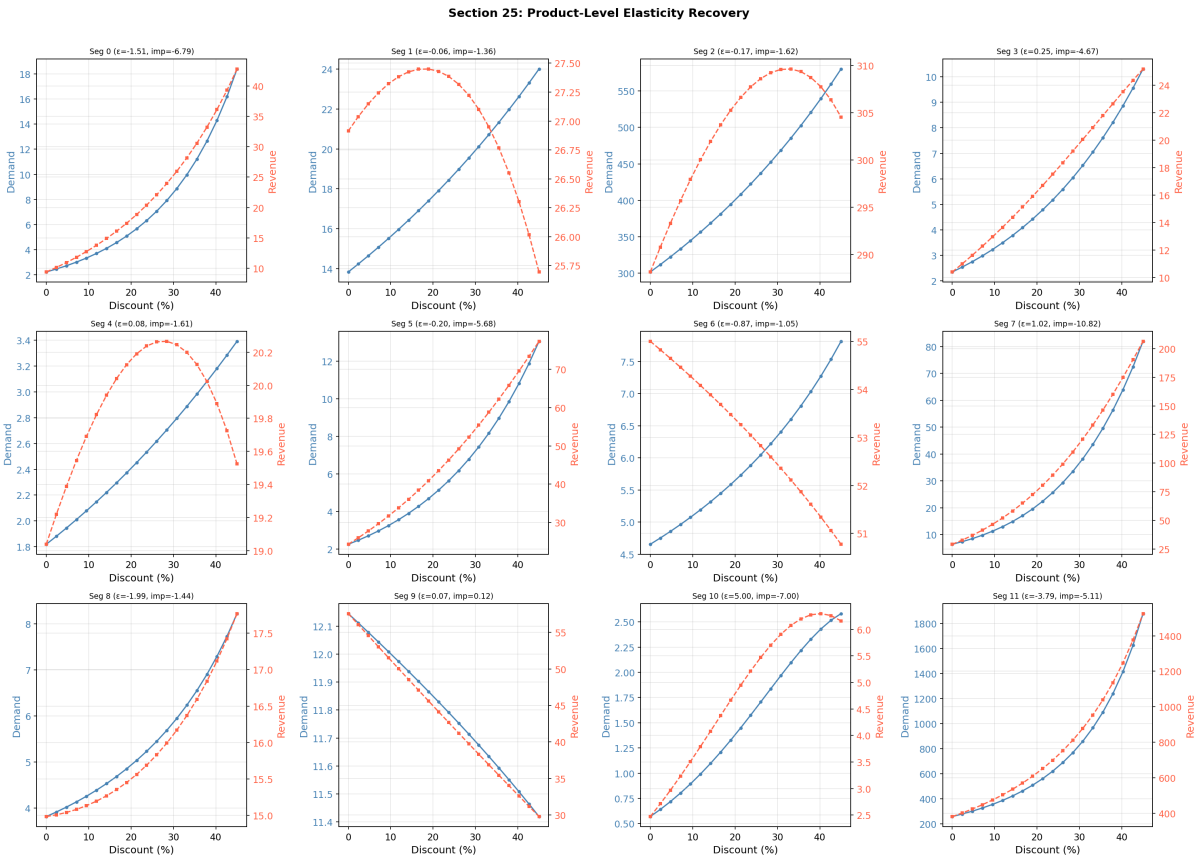

Figure 12: Distribution of shrunk elasticities by segment, showing substantial heterogeneity even within segments.

10 Discussion

10.1 Methodological Contributions

1. **Proper demand specification.** We identify and correct a common error in markdown simulators where the discount effect is double-counted by computing an "effective price" and feeding it to the price elasticity coefficient. When the demand model is estimated with shelf price and discount depth as separate regressors, the simulator must use the shelf price (fixed) for the elasticity term and the discount depth (variable) for the promotional effect term.
2. **AR(1) demand persistence.** The lagged log-demand term $\hat{\rho}_i \ln Q_{i,t-1}$ makes the simulator's demand transition genuinely dynamic: the state variable q_t now causally influences the next-period demand, not just observationally. This addresses the "memoryless transitions" criticism—the simulator's demand process is no longer a repeated static response but an autoregressive process where demand shocks persist over time.
3. **Within-segment cross-product substitution.** The substitution term $\hat{\lambda}_i \overline{d}_{-i,s(i),t}$ captures promotional cannibalization: when competitors in the same segment are discounted, own demand falls. This means the simulator no longer treats products independently—the discount decision for one product affects demand for others, which is critical for realistic policy evaluation of broad vs. targeted promotions.
4. **Stratified representative selection.** Revenue-tercile stratified sampling (50%

top, 30% middle, 20% bottom) within each segment ensures the selected products span the full revenue distribution, rather than being dominated by head SKUs. This reduces head-SKU selection bias and produces more representative estimates of the full product assortment's promotional response.

5. **Full discount accounting.** All three discount components in the Dunnhumby data (retail discount, coupon discount, and coupon-match discount) are included in both the reconstructed shelf price and the discount depth, ensuring consistent price measurement across the estimation and simulation stages.
6. **Continuous promotional intensity.** By using the continuous store-fraction of display and mailer exposure rather than binary indicators, the demand model captures the intensity dimension of promotional activity. A product displayed in 10% of stores receives a proportionally smaller estimated effect than one displayed in all stores.
7. **Seasonality in the simulator.** The simulator includes calendar-aware demand variation via cosine/sine Fourier terms entered as deviations from training-period means. This allows the simulator to evaluate whether promotions deployed early vs. late in a seasonal cycle produce different revenue outcomes.
8. **Parameter-specific shrinkage.** The James–Stein shrinkage weights use HC1 standard errors for each parameter rather than a generic residual-variance proxy, aligning the degree of regularization with the actual estimation uncertainty for each coefficient.
9. **Comprehensive shrinkage scope.** All six slope parameters (elasticity, discount effect, persistence, substitution, display, and mailer) are shrunk toward segment medians, with post-shrinkage sign constraints for economic consistency.

10.2 Limitations

1. **Limited demand persistence.** The AR(1) lagged-demand term captures first-order demand "stickiness" (mean shrunk $\hat{\rho} = 0.039$, 64/150 products nonzero), making transitions no longer memoryless. However, the model does not capture richer intertemporal dynamics such as stockpiling, post-promotion dips, habituation, or reference-price adaptation. These would require higher-order autoregressive terms, regime-switching, or structural models of consumer inventory behaviour.
2. **Linear substitution.** The within-segment substitution term captures first-order cannibalization via the mean discount depth of segment competitors (mean shrunk $\hat{\lambda} = -0.852$, 80/150 products nonzero). However, the deviation-from-mean specification is linear and symmetric. Non-linear patterns (e.g., asymmetric cannibalization where premium products steal share from economy products more than vice versa), cross-segment substitution, and complementarity effects remain outside the model's scope.
3. **Stratified sampling limitations.** The 150 representative products are selected via stratified revenue-tercile sampling (50% top / 30% middle / 20% bottom within each segment), which ensures coverage across the revenue distribution but is not random sampling. The results should be interpreted as characterizing a commercially diverse subset of the assortment, not the full product catalog.
4. **Endogeneity.** Price and discount decisions are made by the retailer based on (partially) observed demand signals. Without instrumental variables, the elasticity estimates suffer from simultaneity bias. We partially mitigate this by separating

shelf-price elasticity from promotional discount effects.

5. **Elasticity attenuation.** The median shrunk elasticity of -0.215 is an order of magnitude below the meta-analytic literature mean of -2.62 (Bijmolt et al. 2005). The remaining attenuation toward zero is a known consequence of limited shelf-price variation in the data and the non-positivity constraint.
6. **ML models not structurally comparable.** The tree-based and neural network benchmarks are trained as flexible predictors and then collapsed to two-parameter log-linear approximations via finite differences for simulator integration. This is not equivalent to simulating the original ML model. The selection of OLS for the simulator is a structural compatibility decision, not solely a performance comparison.
7. **Post-shrinkage constraints are modeling choices.** Winsorization at the 2.5th/97.5th percentiles, the non-positivity constraint on elasticities, the [0, 0.95] stationarity constraint on persistence, and the non-positivity constraint on substitution are ex post structural edits that make the simulator stable for optimization, but less faithful as a pure measurement device.
8. **Independent promotional noise.** Display and mailer events are drawn independently of the retailer's discount decision. In practice, the retailer jointly decides on discounts and promotional support (display, mailer), which creates selection effects that our model does not capture.

11 Conclusion

We have developed a calibrated product-level promotion response simulator that:

1. Estimates demand at the individual SKU level using log-linear models with continuous promotional controls (display fraction, mailer fraction), seasonality, AR(1) demand persistence, and within-segment cross-product substitution, achieving 46.1% out-of-sample MAPE and 0.694 lift correlation.
2. Properly separates shelf-price elasticity from promotional discount effects in the simulator demand specification, avoiding the common double-counting error.
3. Applies James–Stein empirical Bayes shrinkage with parameter-specific standard errors and post-shrinkage constraints to produce stable demand parameters for 150 representative products (selected via stratified revenue-tercile sampling) from 11 segments, shrinking six coefficients—elasticity, discount effect, display effect, mailer effect, demand persistence, and substitution effect—toward segment medians.
4. Incorporates calendar-aware seasonality and stochastic promotional events (display and mailer draws from truncated normal distributions) as deviations from training-period means, enabling evaluation of temporally varying promotion strategies.
5. Identifies targeted discount strategies that improve revenue by up to 41.3% over the no-discount baseline, with Monte Carlo confidence intervals confirming result robustness.
6. Reveals that 64.7% of products benefit from discounting, with a median revenue uplift of 105.1% (IQR: 36.8%–288.8%), but the optimal discount depth varies substantially across products and segments—motivating product-level rather than category-level markdown decisions.

The simulator captures own-product promotional response enriched by first-order demand persistence and within-segment substitution under budget constraints. It can serve as the foundation for reinforcement learning or dynamic programming approaches to op-

timal discount scheduling. Extensions to higher-order intertemporal dynamics (stockpiling, post-promotion dips, reference-price effects), cross-segment substitution, and joint discount-display optimization remain important directions for future work.

References

[1] Ban, G.-Y. and Rudin, C. (2019). "The Big Data Newsvendor: Practical Insights from Machine Learning." *Operations Research*, 67(1), 90–108.

[2] Bijmolt, T.H.A., van Heerde, H.J., and Pieters, R.G.M. (2005). "New Empirical Generalizations on the Determinants of Price Elasticity." *Journal of Marketing Research*, 42(2), 141–156.

[3] Blattberg, R.C. and Neslin, S.A. (1990). *Sales Promotion: Concepts, Methods, and Strategies.* Prentice Hall.

[4] Breiman, L. (2001). "Random Forests." *Machine Learning*, 45(1), 5–32.

[5] Chen, T. and Guestrin, C. (2016). "XGBoost: A Scalable Tree Boosting System." *Proc. KDD*, 785–794.

[6] Duan, N. (1983). "Smearing Estimate: A Nonparametric Retransformation Method." *Journal of the American Statistical Association*, 78(383), 605–610.

[7] Efron, B. and Morris, C. (1975). "Data Analysis Using Stein's Estimator and its Generalizations." *Journal of the American Statistical Association*, 70(350), 311–319.

[8] Ferreira, K.J., Lee, B.H.A., and Simchi-Levi, D. (2016). "Analytics for an Online Retailer: Demand Forecasting and Price Optimization." *Manufacturing & Service Operations Management*, 18(1), 69–88.

[9] Ke, G. et al. (2017). "LightGBM: A Highly Efficient Gradient Boosting Decision Tree." *Advances in Neural Information Processing Systems*, 30.

[10] Phillips, R.L. (2005). *Pricing and Revenue Optimization.* Stanford University Press.

[11] Talluri, K.T. and van Ryzin, G.J. (2004). *The Theory and Practice of Revenue Management.* Springer.

[12] Tellis, G.J. (1988). "The Price Elasticity of Selective Demand: A Meta-Analysis of Econometric Models of Sales." *Journal of Marketing Research*, 25(4), 331–341.

[13] Van Heerde, H.J., Leeflang, P.S.H., and Wittink, D.R. (2004). "Decomposing the Sales Promotion Bump with Store Data." *Marketing Science*, 23(3), 317–334.